

Connecting PCA & SVD with Math

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PCA gives us a coordinate system based on directions in our data.

Remember our data matrix $\underline{X}$ from the SVD content, but this time with our units being row vectors:

$$\underline{X} = \begin{bmatrix} \text{---} x_1 \text{---} \\ \text{---} x_2 \text{---} \\ \vdots \\ \text{---} \end{bmatrix}_n$$

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So now rows would ~~the~~ correspond to a person (in a demographic dataset) or a house (in a housing price dataset).

First, we will compute the average "unit", taking $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$.

We then use this $\bar{x}$ to create an average matrix:

$$\underline{\bar{X}} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \begin{bmatrix} \text{---} \bar{x} \text{---} \end{bmatrix}$$

Finally, we can "mean-center" ~~our~~ our data by taking ~~the~~ $A = \underline{X} - \underline{\bar{X}}$.

Connection to material on the slides: this means A is centered at the origin!

The next math step of PCA is to calculate the covariance matrix of our mean-centered data, which we will denote as $C = A^T A$.

Let us recall from SVD that we can break down/decompose A into:

$$A = U \Sigma V^T$$

Remember that what we want from PCA is the directions our data has highest variance in. To find this, it makes sense to ~~calculate~~ calculate the covariance matrix of A , which we will denote C .

$$C = A^T A$$

From linear algebra, we know the eigenvectors of C ~~gives~~ give us a basis that informs us about the spread of the data. This should sound very similar to the singular vectors of X ! So let us find these eigenvectors & their associated eigenvalues. E.g. we have the first eigenvector $v_1^T A^T A v_1$, etc. Putting these eigenvectors in a matrix called V (same as from SVD!) gives us that our principal components (denoted as P) are equal to the amount of ^{the} variance in our dataset that the loadings capture.

Thus, we have

$$P = AV$$

Diagram annotations for $P = AV$:

- P : principal components
- A : mean-centered data
- V : singular vectors
- Arrows from V to P : loadings
- Arrows from A to P : singular values

In the SVD lens, we also have that $P = U \Sigma$, since this also represents ~~how much~~ the direction our data varies in and by how much.

Big takeaway: you can get PCs AND loadings directly from SVD $\rightarrow$ it's a very powerful tool!

Lastly, we will discuss a further interpretation of the singular values as a way to determine the proportion of variance explained by each PC.

The eigenvalues λ of V are equal to the square of the singular values: $\lambda_i = \sigma_i^2$. So to figure out how much variance is being captured in the first k PCs, we can take

$$\frac{\sum_{j=1}^k \lambda_j}{\sum_{i=1}^n \lambda_i} = \text{P.V.E. of first } k \text{ PCs}$$

So if I want to get a lower-dimensional representation of my data that still captures most of the signal in the data, I can keep only the amount of PCs needed to explain 90% (or some percentage ~~that~~ that is user-selected) of the variation in my data.